

MamaCare AI - 10-Minute Pitch Script

Slide 1 - The Team (30 seconds)

Quick intro. Don't dwell. Names + credibility.

"We are Kasalite. Our team combines public health, AI engineering, health innovation, and business analytics. We built MamaCare AI for the women who fall through the cracks of maternal health systems."

Slide 2 - Problem and Context (2 minutes)

Set the stage. Let the weight of the numbers land. Then pivot to the gap.

"Every day, 800 women die from preventable pregnancy-related causes. That is WHO's number. Not 800 a year. 800 a day. And 75% of those deaths are preventable.

The question is: why?

The answer is not that we don't know how to save them. We do. The answer is that care stops at the clinic door. A woman visits, gets checked, goes home - and what happens between visits is invisible to the health system.

Now most digital health solutions assume internet access and smartphones. But in rural Sub-Saharan Africa, only one in three women has mobile internet. The majority have a basic phone. Voice calls and SMS. Nothing more.

Meet Abena. She represents millions of women. She has no app. No data plan. No way to call for help when a symptom starts at 2 AM. Her next appointment is weeks away. Between now and then, anything could happen.

That gap is where maternal deaths happen. And that is exactly what we built MamaCare AI to fill."

Slide 3 - Solution (2 minutes)

Contrast with the problem. Describe how it works in simple terms.

"MamaCare AI is a voice-based agent that calls pregnant women on their basic phones, in their language, and checks how they are doing.

There is no app. No smartphone. No data. The woman just picks up when the phone rings.

Here is how it works:

The system calls her on a regular schedule tied to her stage of pregnancy. It asks her questions - how she is feeling, any symptoms, any changes. She answers in her own words. The AI listens, assesses, and responds.

If she is fine - LOW risk - it gives her routine advice and schedules the next call.

If something seems off - MEDIUM risk - it flags her case for the health worker to review within 24 hours.

If there is a danger sign - HIGH risk - an alert fires immediately to the clinic. The midwife sees it, calls the mother, and can escalate to a referral.

And this does not stop at pregnancy. When a pregnancy ends in loss, MamaCare does not go silent. The mother is moved into a post-loss support track - check-ins, mental wellness support, physical recovery monitoring.

The key principle: zero friction. No barrier to access. A basic phone and a voice call are all it takes."

Slide 4 - Clinical Dashboard (1.5 minutes)

Bridge from the patient side to the health worker side.

"For the health worker, we built a dashboard. This is the other half of the system.

When a mother reports symptoms on a call, the result appears here in real time. The dashboard has three modules:

First, the Risk Overview - a live feed of cases sorted by urgency. HIGH risk cases appear at the top instantly, with a WebSocket alert that pushes to the screen without any page refresh.

Second, User Management - an activity log of every patient interaction. Who was called, what was reported, what action was taken.

Third, Health Alerts - the ability to send personalized messages to individual patients - a medication reminder - or mass alerts to all patients - for example, a malaria outbreak notification.

The dashboard works on any device - a desktop at the clinic, a tablet in the field, or a basic smartphone. No installation needed.

Together, the voice calls and the dashboard form a complete loop: patient reports symptom → AI triages → dashboard alerts → clinician acts."

Slide 5 - Methodology, Ethical and Responsible AI (1.5 minutes)

This is the credibility slide. Be honest about the numbers but confident about the approach.

"Let me walk you through how the AI works.

The core method is called Guideline-Driven Few-Shot Fine-Tuning, or GDF-FT. The 'guideline' part is key - we did not train this model on random internet data. We trained it on Ghana's National Safe Motherhood Protocol, the official clinical guidelines from the Ghana Health Service.

We started with 28 clinical anchor phrases extracted from the protocol. That was our seed data. We balanced it through oversampling and used it to fine-tune a compact multilingual AI model called DistilBERT.

The result? 66.67% overall accuracy.

Let me break that down honestly:

- For HIGH risk cases, the model achieved a recall of 1.0 - it caught every single danger case. When a woman reported something life-threatening, the system did not miss it.
- For LOW risk, precision and recall were 0.80 - solid and balanced.
- MEDIUM risk is where we need to improve - a recall of 0.20 means we miss most medium cases. They get classified as LOW instead.

Why present this honestly? Because we know the model needs work. The Technical Document itself says: 'We acknowledge that more work needs to be done.' But catching every HIGH risk case with just

28 seed examples is a proof point for the methodology.

Note: The slide includes **"Ethical and Responsible AI"** in its title but does not yet contain that content. The team should add ethical AI details from the Technical Document if the slide needs them."

Slide 6 - Data Use, Open Science and Reproducibility (1 minute)

Shorter slide. Emphasize transparency and openness.

"Everything we built sits on open foundations.

The clinical guidelines we used - the National Safe Motherhood Protocol - are publicly available from the Ghana Ministry of Health. The AI model, DistilBERT, is open source from HuggingFace. Our dashboard stack - React, Express.js, PostgreSQL - is entirely open source. And all our code is on GitHub under the MIT License.

This means anyone can:

- Verify our results by running the same code
- Adapt MamaCare for their own context or language
- Build on our work without asking permission

In research, reproducibility matters. We made sure our project scores well on that front."

Slide 7 - Target and Value Proposition (1 minute)

Show we understand all stakeholders.

"Three groups benefit from MamaCare AI:

First, the mother. She gets regular voice check-ins with no app, no data, no smartphone needed. Continuous monitoring that catches things between clinic visits.

Second, health facilities. The AI triages routine cases automatically, so the midwife is not drowning in low-risk follow-ups. Digital records replace the paper register.

Third, district health administration. They get real-time KPIs - how many mothers enrolled, how many HIGH risk cases detected, how fast they were acted on. Exportable reports for planning and budgeting.

Each group gets exactly what they need. The mother gets care. The clinic gets efficiency. The government gets data."

Slide 8 - Impact (1 minute)

Connect to the bigger picture. SDGs judges care about.

"The impact aligns with three Sustainable Development Goals.

SDG 3 - Good Health and Well-Being. MamaCare directly reduces maternal mortality by catching danger signs between clinic visits before they become fatal.

SDG 5 - Gender Equality. Maternal mortality is a gender issue. Women die because health systems fail to reach them. MamaCare removes the access barrier - no app, no internet, no literacy required.

SDG 10 - Reduced Inequalities. Poor, rural women die at far higher rates than wealthy urban women. MamaCare works on a basic phone over GSM, which reaches over 90% of the population. No infrastructure gap.

Three concrete outcomes:

- Reduced maternal mortality through continuous monitoring
- Increased health system productivity - AI handles the routine, human clinicians handle the critical
- Zero infrastructure barrier - if there is a GSM signal, MamaCare works"

Slide 9 - Next Steps: Scalability and Sustainability (30 seconds)

Close with momentum. Show we have a plan.

"Three next steps:

One - expand language support to Twi, Fante, Ga, and Dagbani, Ghana's four main local languages.

Two - improve the model. More guideline anchors from the Safe Motherhood Protocol, new training strategies, a larger dataset. We know 66.67% is the floor, not the ceiling.

Three - integrate into the Ghana Health System. Embed MamaCare into existing CHPS compounds and district health structures. Not a parallel system - a tool that makes the existing system better.

This is not a prototype looking for a problem. This is a deployed system that works on Railway, ready for the next phase. What we need now is partnership and field validation to take it from proof of concept to national impact."